

PBSMS

Frontend Design Specification

Version 1.0 · Companion to SRS v2.1, Chapter 34

Complete interface proposal across all system modules

Item	Value
Document	PBSMS Frontend Design Specification
Version	1.0 — Proposed
Status	For review. No part of this has been user-tested; see §16.
Governs	SRS v2.1 Chapter 34 (Frontend Architecture, UX & Offline/PWA Strategy) and Chapter 42 (Accessibility)
Scope	All modules across SRS Volumes 0–V, expressed as four frontend surfaces
Companion prototype	pbsms-frontend-prototype.html — interactive, self-contained

1. Purpose & Relationship to the SRS

SRS v2.1 Chapter 34 specifies frontend behaviour — an application shell, a design-token layer, offline capability for attendance and score entry, WCAG 2.1 AA, and distinct UI states. It does not specify a single screen. This document fills that gap and nothing else: it proposes the interface, and it defers to the SRS on every functional question.

Where this document and the SRS disagree, the SRS wins. Where this document proposes something the SRS does not cover, it is flagged as a proposal, not a requirement.

1.1 What this document contains

- A design-token system with real values (§4) — Chapter 34.2 requires tokens but defines none.
- Four frontend surfaces (§3), because 'the frontend' is not one product.
- A complete screen inventory across every module in SRS Volumes 0–V (§8).
- Detailed specifications for the ten screens that carry the most risk (§9).
- Offline and sync UX, satisfying FR-UX-020 and FR-UX-030 (§10).
- Role-based visibility rules mapped to Chapter 13 scope (§11).
- Accessibility implementation detail against WCAG 2.1 AA (§12).
- A performance budget tied to NFR-PERF-020 (§13).

2. Design Principles

These are derived from constraints in the SRS and in the market, not from general design preference. Each one is falsifiable — you can tell whether a screen obeys it.

Principle	Derived from	What it forbids
The register is the product	Attendance and score entry are the highest-frequency actions in the system (FR-ATT-010, FR-ASM-020)	Burying score entry three levels deep behind an academics menu. A teacher reaches either in one tap from their home screen.
Never lie about sync	FR-UX-030	Optimistic checkmarks. A row is 'saved on this phone' or 'saved to the school' and the interface always says which.
Money is auditable on screen	Chapter 23.1 — every balance derived from valid transactions	A balance shown as a single number with no path to the transactions that produced it.
Published is different from draft	FR-RES-030 — published results are never overwritten	Any visual treatment where a draft result and a published result look alike.
Data before decoration	NFR-PERF-020 (2.5s on 3G), FR-UX-040 (low data)	Hero images, icon fonts, decorative illustration, webfont stacks over 100KB.
One tenant at a time	TEN-012, TEN-013	Any screen that mixes two tenants' data, including in platform support tooling.

3. The Four Surfaces

The single most consequential proposal in this document: PBSMS is not one interface. It is four, sharing one design system, one component library and one codebase. Designing them as one app produces a desktop admin tool that teachers cannot use on a phone in a classroom — which is the failure mode that kills school software adoption.

Surface	Primary user	Device reality	Design posture
Staff Console	Head, administrator, accountant, coordinator, exam officer, admissions officer	Laptop or shared desktop, office, reliable power	Data-dense. Tables, filters, bulk actions, keyboard shortcuts. Desktop-first, responsive down.
Teacher Field App	Class and subject teachers	Personal Android phone, in a classroom, connectivity unreliable, one hand free	Offline-first. One task per screen, large touch targets, thumb-zone actions, works fully offline.
Parent View	Guardians	Low-cost Android, metered data, often arrives from a WhatsApp link	Read-mostly, extremely light. No app install required. Loads under 2.5s on 3G or it has failed.
Platform Console	PBSMS staff (Chapter 3 roles)	Laptop, internal	Deliberately austere. Tenant-scoped, audit-visible, no bulk destructive actions.

These share: tokens, components, auth, and the offline/sync layer. They do not share navigation, information density, or layout. A parent never sees a sidebar; a teacher never sees a data table.

4. Design Tokens

Chapter 34.2 requires 'central design tokens' governing colour, typography, spacing and breakpoints, with a per-tenant branding layer on top. Here they are. Every value below is a proposal open to change — but the system needs concrete values, and 'we will decide later' is how design debt starts.

4.1 Colour — rationale

The palette comes from the subject's own materials: the deep green-black of a classroom blackboard, the warm manila of an exercise book, and the gold that already signals attainment in Ghanaian schools and on the national flag. It is deliberately not a generic SaaS blue.

Gold is reserved. It marks attainment and nothing else — a passing grade, a completed term, a settled account. If gold starts appearing on buttons and headers it stops meaning anything, and the one visual signal a parent scanning a report card actually needs is lost.

Core

Token	Value	Use	Contrast on canvas
--pb-ink	#12211F	Body text, headings	14.8:1 — AAA
--pb-ink-muted	#5A6663	Secondary text, table meta, captions	5.4:1 — AA
--pb-teal-900	#0B4F4A	Primary actions, headers, active nav	9.1:1 — AAA
--pb-teal-600	#12726B	Hover, links, focus ring	5.9:1 — AA
--pb-teal-50	#E6EFEC	Selected rows, subtle fills	—
--pb-gold	#9A6700	Attainment only — grades, settled, completed	5.2:1 — AA
--pb-gold-50	#FBF3E0	Attainment fills, award chips	—
--pb-canvas	#FAF7F0	Page background (warm, paper)	—
--pb-surface	#FFFFFF	Cards, tables, sheets	—
--pb-rule	#E3DED2	Hairlines, table borders, dividers	—

Every contrast ratio in this table is calculated, not tool-verified. Re-check each one against a contrast checker before implementation — see §11.2 and §14.

Semantic

Token	Value	Meaning in this product
--pb-success	#1E6B3A	Synced, posted, approved, present
--pb-warning	#8A5A00	Pending review, overdue soon, unsynced
--pb-danger	#A32017	Failed, reversed, absent, overdue, validation error
--pb-info	#12726B	Neutral system state

Colour is never the only carrier of meaning (WCAG 1.4.1). Every state above pairs with a shape, an icon, or text — see §12.1.

4.2 Typography

Two faces, one of them free. The body face is the device's own system stack, which costs zero bytes and renders familiarly on cheap Android — a direct concession to FR-UX-040. One display face is loaded, subset, for headings and figures.

Role	Face	Why
Display / headings	Archivo (variable, subset to Latin, woff2, ~28KB)	A grotesque with enough width and weight range to carry dense admin headers without a second face. Not a high-contrast serif — those read as editorial, and this is a working tool.
Body / UI	system-ui stack — Roboto on Android, SF on iOS, Segoe on Windows	Zero network cost, instantly familiar, no FOUT on the screens teachers use most.

Role	Face	Why
Figures & identifiers	Archivo with font-variant-numeric: tabular-nums; identifiers in ui-monospace	Scores, money and attendance counts must align in columns. Admission numbers, invoice numbers and BECE index numbers are read aloud and cross-checked against paper — monospace makes transcription errors visible.

Type scale (1.200 minor third, 16px base)

Token	Size	Weight	Use
--pb-text-2xs	11px	500	Table meta, timestamps
--pb-text-xs	12px	500	Labels, chips, captions
--pb-text-sm	14px	400	Table cells, secondary body
--pb-text-base	16px	400	Body. Never smaller for parent-facing text.
--pb-text-lg	19px	500	Card titles, section heads
--pb-text-xl	23px	600	Page titles
--pb-text-2xl	28px	600	Dashboard figures
--pb-text-3xl	33px	700	Teacher app primary figures, parent balance

4.3 Space, radius, elevation, motion

Group	Tokens
Space (4px base)	--pb-space-1 4px · -2 8px · -3 12px · -4 16px · -5 24px · -6 32px · -7 48px · -8 64px
Radius	--pb-radius-sm 4px (inputs, chips) · -md 8px (cards, sheets) · -lg 12px (modals) · -full (avatars only)
Elevation	--pb-shadow-1 0 1px 2px rgba(18,33,31,.08) · -2 0 4px 12px rgba(18,33,31,.10) · -3 0 12px 32px rgba(18,33,31,.14). Three levels only; anything needing a fourth is a layout problem.
Motion	--pb-ease standard cubic-bezier(.2,0,0,1) · --pb-dur-fast 120ms (state) · -base 200ms (sheets, drawers) · -slow 320ms (page transitions). All motion suppressed under prefers-reduced-motion.
Breakpoints	--pb-bp-sm 480px · -md 768px · -lg 1024px · -xl 1280px. Teacher app and Parent view are designed at 360px first — the real floor for Android in this market.

4.4 Tenant branding layer

Per Volume 0 Chapter 6, branding is per tenant. Exactly three things are tenant-overridable, resolved once at the layout root and never per screen:

- --pb-brand-primary — replaces teal in nav, primary buttons and document letterheads. Validated on upload for 4.5:1 contrast against white; rejected with a clear message if it fails, because a tenant must not be able to configure their own school into an inaccessible interface.
- Logo — used in the shell header, report cards, receipts and certificates.
- School display name and letterhead block.

Everything else — type, spacing, semantic colour, component shape — is fixed. Tenants get identity, not a theme engine. This keeps the component library testable and stops a school from breaking its own accessibility conformance.

5. Component Library

Built once, used across all four surfaces. Naming follows what users control, not how it is built (a person manages notifications, not webhook config).

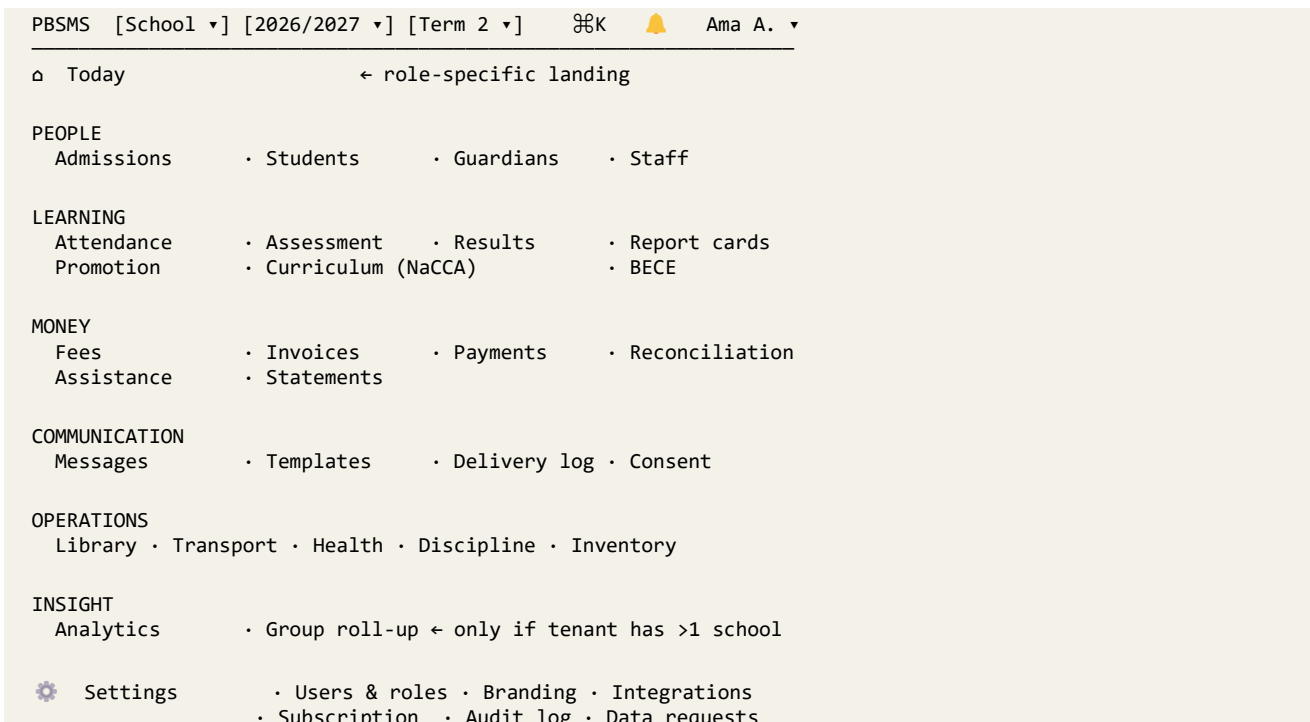
Group	Components
Foundational	Button (primary/secondary/ghost/danger), IconButton, Link, Chip, Badge, Avatar, Tag, Divider, Spinner, ProgressBar
Form	TextField, NumberField (tabular figures), Select, Combobox, DatePicker (term-aware), Checkbox, Radio, Toggle, Textarea, FileUpload, ScoreInput (bounded, keyboard-fast), CurrencyInput (GHS-aware), SearchField
Data	DataTable (sortable, filterable, paginated, column-configurable), DefinitionList, KeyFigure, TrendSparkline, EmptyState, StatusPill, AuditTrail, Timeline, ComparisonBar
Navigation	AppShell, Sidebar, TopBar, ContextSwitcher (school/campus/year/term), Breadcrumbs, Tabs, Stepper, BottomNav (mobile), CommandPalette
Feedback	Toast, InlineAlert, ConfirmDialog, DestructiveConfirm (types-to-confirm), FieldError, FormSummaryError, OfflineBanner, SyncLedger
Surface	Card, Sheet (mobile bottom sheet), Modal, Drawer, Popover, Tooltip, Accordion
Domain	RegisterRow, ScoreGrid, ResultCard, GradeChip, InvoiceLine, PaymentAllocator, BalanceSummary, ReportCardPreview, CompetencyMeter, ApprovalBar, ConsentToggle

The Domain group is where this product differs from any generic admin template. A ScoreGrid that understands assessment weighting, a PaymentAllocator that splits one payment across invoice lines, and a CompetencyMeter that renders NaCCA indicator coverage cannot be bought off the shelf and should be built with the most care.

6. Navigation & Information Architecture

6.1 Staff Console

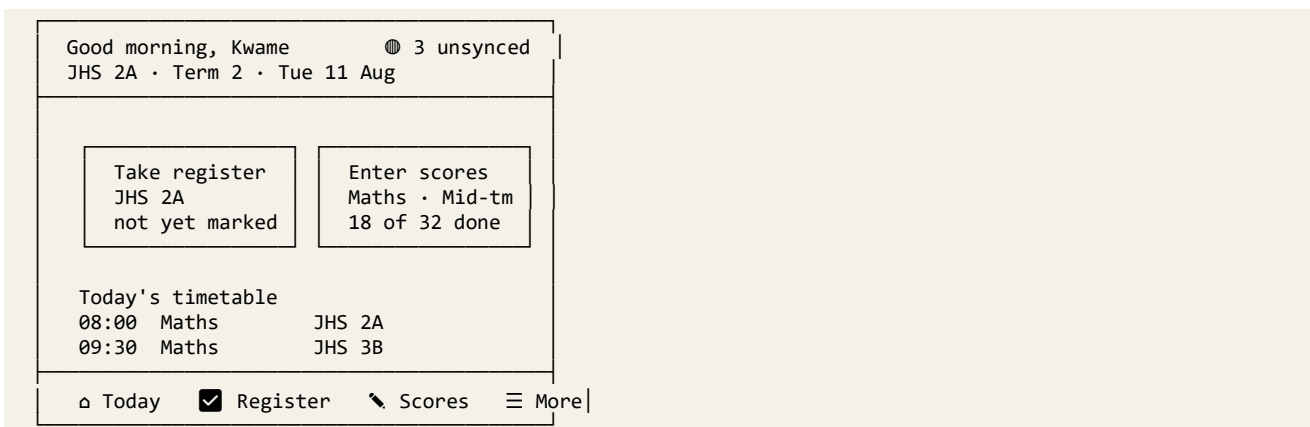
Sidebar grouped by how school staff actually think about their work, not by SRS volume order. Groups collapse; items are permission-generated (Chapter 34.1).



The context switcher (school / academic year / term) sits in the top bar and is global. Chapter 34.1 requires the active year, term and campus to be visible at all times — the single most common cause of data-entry error in school software is a teacher entering scores into last term.

6.2 Teacher Field App

No sidebar. Four bottom-nav destinations, sized for one-handed use, with the two highest-frequency tasks reachable in one tap from launch.



6.3 Parent View

No navigation chrome at all on first load. A parent arriving from a WhatsApp link sees the thing they were notified about, immediately, with everything else one tap below it. Multiple children appear as a horizontal switcher, not a menu.

6.4 Platform Console

Tenant list → tenant detail. Every screen states which tenant is in view, in a persistent bar that is visually distinct from the tenant-facing product (dark, unmistakable) so PBSMS staff can never confuse the console with a customer's own account. Impersonation adds a second, louder banner naming the ticket and the expiry time (TEN-021).

7. Screen Inventory — All Modules

Complete coverage of SRS Volumes 0–V. Surface key: S = Staff Console, T = Teacher Field App, P = Parent View, X = Platform Console.

7.1 Platform & Tenancy (Volume 0)

Screen	Surface	SRS reference
Tenant list (search, status filter, plan, student count)	X	Ch 2, Ch 4
Tenant detail — profile, schools, subscription, health	X	Ch 2, Ch 5
Tenant lifecycle actions — suspend, offboard, reinstate	X	Ch 4.1
Plan & pricing configuration	X	Ch 5.1
Metering & usage — active students per tenant per period	X	TEN-030, TEN-031
Platform invoices & dunning status	X	FR-BIL-010..040
Impersonation request & active-grant banner	X	TEN-021
Platform audit log	X	TEN-022
Platform revenue dashboard (MRR, churn, ARPU)	X	FR-BIL-050

7.2 Onboarding & Tenant Settings

Screen	Surface	SRS reference
Signup / trial start	S	Ch 4.1
Onboarding wizard — school, year, terms, divisions, levels	S	FR-ONB-010
Starter data review (grading scale, fee categories, templates)	S	FR-ONB-010
Bulk import — upload, map columns, dry-run validation, commit	S	FR-ONB-020
Import error report & re-upload	S	FR-ONB-020
Branding — logo, primary colour, letterhead	S	Ch 6, Ch 34.4
Users & roles, delegation, conflict-of-interest warnings	S	Ch 13
Integrations — payment, messaging, credential health	S	Ch 36
Subscription & billing (tenant side)	S	TEN-031

7.3 Dashboards (Chapter 14)

Screen	Surface	SRS reference
Proprietor dashboard — group roll-up across schools	S	FR-ANL-010
Headmaster dashboard — approvals, attendance, collections	S	Ch 14.1
Accountant dashboard — today's collections, unverified, overdue	S	FR-FIN-040
Teacher home — register + scores + timetable	T	Ch 14.1
Parent home — child summary, balance, latest results	P	FR-STU-060
Student home — own results, timetable	P	FR-RES-040
KPI configuration	S	Ch 14.2
Action tracker & escalations	S	FR-COM-060

7.4 Admissions (Chapter 15)

Screen	Surface	SRS reference
Application list — state filter, ageing, source	S	FR-ADM-010
Application detail — identity, guardians, documents, medical	S	FR-ADM-010

Screen	Surface	SRS reference
Duplicate-match review (side-by-side, merge or dismiss)	S	FR-ADM-020
Interview / entrance assessment scheduling & outcome	S	FR-ADM-010
Approval & conversion to student (atomic, confirm step)	S	FR-ADM-030
Admissions KPIs — volume, processing time, conversion	S	FR-ADM-050

7.5 Students & Guardians (Chapter 16)

Screen	Surface	SRS reference
Student list — filter by class, status, division, campus	S	FR-STU-010
Student profile shell with tabs (below)	S	FR-STU-010
· Identity & enrolment history	S	FR-STU-010
· Guardians & relationship flags	S	FR-STU-020
· Academics — results by term and year	S	FR-RES-050
· Attendance — trend and exceptions	S	FR-ATT-050
· Finance — invoices, payments, balance	S	Ch 24
· Health (restricted access)	S	FR-OPS-030
· Discipline (restricted access)	S	FR-OPS-040
· Documents	S	FR-DOC-010
· Timeline — every significant event, chronological	S	FR-STU-050
Transfer — class, campus, inbound, outbound	S	FR-STU-030
Guardian directory & linked children	S	FR-STU-020

7.6 Academic Structure (Chapter 17)

Screen	Surface	SRS reference
Academic years & terms — states, windows	S	FR-ACA-020
Divisions, class levels, classes	S	FR-ACA-010
Subjects by division/level (+ optional NaCCA mapping)	S	FR-ACA-030, DOM-010
Teacher assignment & workload view	S	FR-ACA-050
Timetable builder with live conflict detection	S	FR-ACA-040
Timetable views — class, teacher, room	S	FR-ACA-040

7.7 Attendance (Chapter 18)

Screen	Surface	SRS reference
Daily register — offline-capable	T	FR-ATT-010
Register (desktop fallback)	S	FR-ATT-020
Correction with reason (retains original)	S	FR-ATT-030
Class & school attendance summary	S	FR-ATT-050
Absence follow-up queue & guardian contact	S	FR-ATT-040
Staff attendance	S	Ch 18.1
Child attendance view	P	FR-STU-060

7.8 Assessment, Grading & Results (Chapters 19–21)

Screen	Surface	SRS reference
Assessment configuration & weighting (must total 100%)	S	FR-ASM-010
Score entry grid — desktop, keyboard-fast	S	FR-ASM-020
Score entry — offline-capable mobile	T	FR-ASM-020, FR-UX-020
Grading scale & policy versions	S	FR-GRA-030, FR-GRA-070
Grading run — preview, blocking errors, no silent defaults	S	FR-GRA-060
Results workflow queue (draft → published)	S	FR-RES-010
Result review & return-with-comment	S	FR-RES-010
Publication gate — checklist of unmet conditions	S	FR-RES-020
Published results & version history	S	FR-RES-030
Reopen with reason & authority	S	FR-RES-030
Result analytics — trend, class, subject, division	S	FR-RES-050
Child results view	P	FR-RES-040

7.9 Documents & Promotion (Chapter 22)

Screen	Surface	SRS reference
Report card batch generation with progress	S	FR-DOC-010, NFR-PERF-023
Report card template editor & preview	S	FR-DOC-030
Certificate issue (JHS completion)	S	FR-PRO-010
Public document verification lookup (QR target)	P	FR-DOC-020
Promotion run — system recommendation vs decision	S	FR-PRO-010, FR-PRO-020
Promotion approval & next-year enrolment creation	S	FR-PRO-030

7.10 Finance (Chapters 23–25)

Screen	Surface	SRS reference
Fee structure builder — year, term, level, category	S	FR-FEE-010
Instalment plan & proration rules	S	FR-FEE-020, FR-FEE-030
Penalty rules	S	FR-FEE-040
Invoice generation run & preview	S	FR-PAY-040
Invoice list & detail with line items	S	Ch 24.1
Payment capture — cash, MoMo, card, transfer	S	FR-PAY-010
Payment allocation across invoice lines	S	Ch 24.1
Receipt issue & reprint	S	Ch 24.4
Financial assistance — scholarship, discount, waiver	S	FR-FIN-010
Reversal request & approval (maker-checker)	S	FR-FIN-020
Reconciliation workspace — provider vs internal	S	FR-FIN-030
Student statement of account	S	Ch 24.1
Finance dashboard	S	FR-FIN-040
Invoice, balance & pay now	P	FR-PAY-010

7.11 Communication (Chapter 26)

Screen	Surface	SRS reference
Compose — channel picker with cost estimate	S	FR-COM-040

Screen	Surface	SRS reference
Template library & version editor	S	FR-COM-020
Audience builder — class, level, arrears, custom	S	FR-COM-040
Send review — recipient count, channel split, cost	S	FR-COM-030
Delivery log — per recipient, per channel, retry state	S	FR-COM-040
Consent registry & withdrawal handling	S	DP-070
Message thread	P	FR-COM-060

7.12 Analytics (Chapter 27)

Screen	Surface	SRS reference
School analytics — attendance, performance, collections	S	FR-ANL-020
Group roll-up comparison across schools	S	FR-ANL-030
Cohort & trend explorer	S	FR-ANL-020
AI-drafted summary with human-approval gate	S	FR-ANL-040

7.13 Supporting Operations (Chapter 28)

Screen	Surface	SRS reference
Library — catalogue, circulation, fines	S	FR-OPS-010
Transport — routes, stops, vehicles, assignments	S	FR-OPS-020
Health — records, incidents, medication log	S	FR-OPS-030
Discipline — case, investigation, response, appeal	S	FR-OPS-040
Inventory — assets, stock, issuance, low-stock alerts	S	FR-OPS-050

7.14 Compliance & Curriculum (Volume V)

Screen	Surface	SRS reference
Consent registry per guardian per channel	S	DP-070
Data subject request queue with 30-day SLA timer	S	DP-030, DP-090
Retention & archive status by record type	S	Ch 40.2
Audit log viewer (tenant-visible, incl. platform actions)	S	TEN-022
NaCCA curriculum mapping — strand/indicator tree	S	DOM-010
Competency coverage report	S	DOM-020
BECE candidate registration & index numbers	S	DOM-040
Mock BECE results & readiness analytics	S	DOM-050, DOM-060
GES statutory return export	S	DOM-070
CSSPS choices & placement outcome	S	DOM-080

8. Key Screen Specifications

Ten screens carry most of the product risk — they are the highest-frequency, the highest-consequence, or the hardest to retrofit. Specified here in detail; the rest follow the patterns they establish.

8.1 Daily Register (Teacher Field App) — the signature screen

Highest-frequency action in the entire system, performed in a classroom, often with no connectivity. Everything about this screen is subordinate to one goal: a teacher marks 32 children in under 45 seconds and knows exactly what has and has not reached the school.

← JHS 2A • Tue 11 Aug	Mark all ▾	
📶 Working offline • 12 marks on this phone	← SyncLedger	
Ama Mensah	[P] A L E S	
✓ on this phone		
Kofi Boateng	P [A] L E S	
✓ saved to school		
Yaa Owusu	P A [L] E S	
✓ on this phone • reason required ⚠		
29 of 32 marked		
[Submit register]		

- Status is a five-state segmented control per row (Present, Absent, Late, Excused, Sick), not a dropdown — one tap, no modal, per FR-ATT-020.
- Each row states its own sync state in words, never a bare icon: 'on this phone' vs 'saved to school'. This is FR-UX-030 taken literally.
- The full roster is pre-fetched when the app last had connectivity, so the screen opens with data even in airplane mode.
- 'Submit register' works offline — it queues. The button never lies: offline it reads 'Submit when back online', and the SyncLedger count increments.
- Rows are 56px minimum, targets 44×44 minimum (WCAG 2.5.5), ordered by surname, sticky search for large classes.

8.2 SyncLedger (component, all Teacher Field App screens)

The proposed signature element of the product. Instead of a spinner, a persistent honest ledger of what has left the device.

📶 Working offline • 12 marks on this phone [View]
 🔄 Sending • 4 of 12 saved to school
 ✓ All 12 saved to school • 2 min ago
 ⚠ 2 could not be saved • [Review]

- Always visible on register and score screens. Never a toast, because a toast disappears and the question 'did my marks save?' does not.
- Tapping it opens the queue: what is pending, what failed, what the failure was, and what to do about it.
- Conflicts (FR-ATT-011) surface here for manual reconciliation when two different users marked the same student — never silently resolved.
- Colour is paired with a distinct glyph and a text label, so the state is legible without colour vision.

8.3 Score Entry Grid (Staff Console)

A spreadsheet-grade editing experience, because that is what it replaces and users will compare it to Excel.

- Keyboard-first: Enter moves down, Tab moves right, values commit on blur, no save button per cell.
- Bounds enforced at the input (FR-ASM-030); an out-of-range value is rejected inline with the valid range stated, not silently clamped.
- Missing scores require an explicit status — absent, exempt, pending — never left ambiguous.

- Weighting bar shows live total; publication is blocked, with the reason stated, while weights do not sum correctly (FR-ASM-010).
- Locked after submission with a visible lock state and a 'request reopen' action rather than a disabled control with no explanation (FR-ASM-040).

8.4 Results Publication Gate (Staff Console)

FR-RES-020 lists five conditions for publication. The screen renders them as a live checklist, not an error after the fact.

Publish • JHS 2A • Term 2 • 2026/2027

- ✓ All 9 subjects approved
- ✓ Class teacher review complete
- ✓ Headmaster approval recorded
- ⚠ 2 students missing Social Studies scores
Ama Mensah • Kofi Boateng [Resolve]
- ✓ Report template available

[Publish] – disabled, with the reason stated

A disabled button with no explanation is the most common failure in approval UI. Here the reason is always adjacent to the control, and every unmet condition links to the screen that resolves it.

8.5 Payment Allocation (Staff Console)

Chapter 24.1's model — payments independent of invoices, joined by allocations — is unusual and must be visible in the interface, or cashiers will misunderstand it.

- Left: the payment (amount, method, reference, provider status). Right: outstanding invoice lines.
- Allocation is explicit — drag or enter amounts per line, with a running unallocated remainder that must reach zero or be deliberately left as credit.
- Unallocated credit is a first-class visible state, not an error.
- Reversal is never a delete. The action reads 'Reverse payment', requires a reason, and produces a linked correcting entry shown in the same timeline (FR-FIN-020).

8.6 Parent Home (Parent View)

Assume arrival from a WhatsApp link on a metered connection. Under 2.5s on 3G, under 150KB total, no login wall for published content reached via an authenticated link.

Sunrise Basic School	
• Ama ○ Kwesi	← child switch
Term 2 results are ready	
[View report card]	
Balance	
GH¢ 450.00 due 30 Aug	
[Pay now]	[See statement]
Attendance this term 96%	
Present 44 • Absent 2 • Late 1	

Note what is absent: no sidebar, no dashboard grid, no KPI cards. A guardian has three questions — how is my child doing, what do I owe, and is anything wrong — and the screen answers them in that order.

8.7 Report Card (Parent View & print)

- One layout that renders on a 360px screen and prints to A4 without a separate template.
- Grade chips use gold for attainment, but the grade is always also stated in text.
- Where the tenant uses NaCCA (DOM-030), the competency profile appears above raw scores, because that is what the curriculum is actually assessing.
- The published version and date are always shown. If a result was later revised, both versions are reachable — FR-RES-030 makes this a data guarantee and the UI must not hide it.

8.8 Reconciliation Workspace (Staff Console)

- Two columns: provider settlement vs internal records, with matched, unmatched and disputed lanes.
- Provider fees are shown as their own line (FR-FIN-030) — a settlement that is short by the fee is not a discrepancy and must not look like one.
- Every unmatched item has one obvious next action, never just a red flag.

8.9 Group Roll-Up (Staff Console, multi-school tenants only)

The capability the old single-school architecture could not express. Appears only when the tenant has more than one school (FR-ANL-030).

- Schools as rows, metrics as columns — collections, attendance, pass rate, outstanding actions.
- Ranked comparison with the tenant's own average as the baseline, not an industry benchmark that does not exist.
- Drill from any cell into that school's own dashboard, with the context switcher following the drill.

8.10 Impersonation Banner (Platform Console)

TEN-021 and TEN-022 require impersonation to be time-boxed, ticket-linked and visible to the tenant in real time. The interface must make it impossible to forget you are inside a customer's account.

■ SUPPORT SESSION • Sunrise Basic School ■
 Ticket #4821 • expires in 47 min • read-only
 This session is visible in the school's audit log.
 [End session now]

Full-width, high-contrast, non-dismissible, present on every screen for the duration. Actions blocked for the platform role (reversal, export, permission change) are visibly disabled with the four-eyes requirement named, not hidden.

9. Offline & Sync UX

Implements FR-UX-020 and FR-UX-030. Scope is deliberate: attendance and score entry are offline-capable. Nothing else is, in v1.

Concern	Behaviour
Pre-fetch	On any successful connection, the teacher's assigned classes, rosters, active assessments and current term config are cached. Cache is tenant-namespaced (TEN-040) and cleared on logout.
Capture	Writes go to IndexedDB first, always. The UI never waits on the network to acknowledge a mark or a score.
Queue visibility	SyncLedger (§8.2) states the count and the state in words at all times on offline-capable screens.
Sync	Background sync on reconnection; manual retry available. Idempotency keys prevent double-posting on retry (NFR-API-010).
Conflict — same user, two devices	Last write by device timestamp wins, silently. Stated in the queue detail so it is discoverable.
Conflict — different users	Surfaced for manual reconciliation, never auto-resolved (FR-ATT-011). Both values, both authors, both times, one decision.
Failure	Named and actionable: what failed, why, what to do. Never a generic retry toast.
Honesty rule	No optimistic UI that implies server persistence. 'Saved on this phone' and 'saved to school' are different sentences and the interface always uses the true one.

10. Role-Based Views

Chapter 13 defines scope from tenant down to record ownership. The interface renders permission-generated navigation (Chapter 34.1) — but per Chapter 34.5, hiding a control is a usability courtesy and never an authorization mechanism. The server decides; the UI merely declutters.

Role	Lands on	Sees	Does not see
Proprietor / Director	Group roll-up if multi-school, else school dashboard	All schools in tenant, finance summary, academic summary	Other tenants. Individual health and discipline detail unless separately granted.
Headmaster	Approvals & exceptions dashboard	Whole school: academics, attendance, finance summary, approvals	Platform settings, other schools unless assigned.
Accountant / Bursar	Finance dashboard	Invoices, payments, allocations, reconciliation, statements	Academic scores, health, discipline.
Academic Coordinator	Results workflow queue	Assessment config, results pipeline, curriculum, analytics	Finance, payroll, health.
Class / Subject Teacher	Teacher Field App home	Own assigned classes and subjects only	Other teachers' classes, finance, admin settings.
Admissions Officer	Application queue	Applications, duplicates, conversion	Enrolled-student finance and academic records beyond conversion.
Parent / Guardian	Parent home	Linked children only; published results, attendance, invoices, messages	Any other child. Unpublished or draft results.
Student	Student home	Own published results and timetable	Finance, discipline, other students.
Platform Support	Tenant list	One tenant at a time, under an active audited grant	Financial reversal, export, permission change without a second approver.

11. Accessibility Implementation

Chapter 42 binds the product to WCAG 2.1 Level AA with a CI gate (NFR-ACC-020) and a pre-GA manual audit (NFR-ACC-030). Naming a standard is not implementing one — this section is what conformance actually requires here.

11.1 Concrete commitments

WCAG criterion	Implementation in PBSMS
1.4.3 Contrast (Minimum)	All token pairs verified at 4.5:1 body / 3:1 large. Tenant brand colours validated on upload and rejected if they fail (§4.4).
1.4.1 Use of Colour	Every status pairs colour with a glyph and a text label. Attendance states, sync states, result states and payment states all readable in greyscale.
2.1.1 Keyboard	Score grid, approval flows, allocation and every dialog fully operable by keyboard. The score grid is designed keyboard-first, not keyboard-compatible.
2.4.7 Focus Visible	2px focus ring in --pb-teal-600 with 2px offset, never removed. Focus is trapped in modals and returned on close.
2.5.5 Target Size	44×44 CSS px minimum on Teacher Field App and Parent View — the surfaces used on phones one-handed.
3.3.1 / 3.3.3 Errors	Errors name the field, state what is wrong and state how to fix it. Form-level summary lists every error with in-page links.
4.1.3 Status Messages	SyncLedger, toasts and validation summaries announce via aria-live so a screen-reader user learns a register synced without moving focus.
1.3.1 Info & Relationships	Data tables use real th/scope. The score grid uses a grid role with correct row and column semantics, not a div soup.

11.2 Verification, not assertion

Every contrast value in §4.1 must be re-checked against a contrast tool before implementation, and the CI accessibility job must fail the build on new critical violations from the first frontend commit — not added once the component library already exists. Retrofitting accessibility across an existing component set is the same category of expensive mistake as retrofitting multi-tenancy across an existing schema.

12. Performance Budget

NFR-PERF-020 requires p95 page load $\leq 2.5s$ on a 3G-equivalent connection. A budget makes that testable per screen rather than aspirational.

Surface	JS (gzipped)	CSS	Fonts	Total first load	Rationale
Parent View	≤ 60 KB	≤ 15 KB	0 KB (system stack only)	≤ 150 KB	Metered data, arrives from a link, must never feel expensive to open.
Teacher Field App	≤ 140 KB	≤ 20 KB	≤ 28 KB	≤ 250 KB	Installed once, cached thereafter; offline shell justifies more.
Staff Console	≤ 250 KB	≤ 30 KB	≤ 28 KB	≤ 400 KB	Desktop, office connection, route-split per module.
Platform Console	≤ 250 KB	≤ 30 KB	≤ 28 KB	≤ 400 KB	Internal use only.

- Route-level code splitting per module; no module's code loads for a user whose role cannot reach it.
- Lists paginate by default (FR-UX-040); no unbounded fetch anywhere.
- Images: student photos lazy-loaded, served at display size, never full-resolution originals in a list.
- Budget enforced in CI as a failing check, alongside the accessibility gate.

13. Build Order

Sequenced so that the two things that cannot be retrofitted — the offline layer and the accessibility baseline — exist before the component library grows.

Stage	Build	Why here
1	Tokens, type scale, base components, focus and contrast baseline, CI a11y gate	Everything else inherits from this. Cheapest moment to get it right.
2	App shell, auth, context switcher, permission-generated navigation	Every screen mounts inside it.
3	Offline/sync layer + SyncLedger, proven with the register	Hardest to retrofit. Build it against one real screen before there are twenty.
4	Teacher Field App — register, score entry	Highest-frequency users; earliest real feedback; validates stage 3.
5	Staff Console — students, academic structure, assessment, results	The operational core.
6	Parent View — home, results, invoices, pay	Unlocks the communication value proposition.
7	Finance console — invoicing, payments, allocation, reconciliation	Depends on 5; highest correctness risk, so not first.
8	Communication, analytics, supporting operations, compliance screens	Breadth after depth.
9	Platform Console	Internal users tolerate rough edges longest; also gated on Chapter 3 impersonation controls existing.

14. Open Questions & Honest Limitations

This specification is a proposal by one designer working from a document, not from users. The following are genuinely unresolved and should not be treated as settled by the confidence of the prose above.

- No user research exists. Nothing here has been shown to a Ghanaian teacher, headmaster, bursar or parent. The register design in §8.1 is the highest-conviction screen and also the one most likely to be wrong in a way only a real teacher would catch. Test it before building stage 4.
- The four-surface split (§3) is the load-bearing structural bet. If it is wrong, most of this document changes. It should be reviewed by whoever holds the Engineering Lead role before stage 1 begins.
- Colour and type are proposals with reasoning, not brand decisions. If PBSMS acquires a brand identity later, §4.1 and §4.2 are the parts to replace — the rest of the document survives that change.
- Contrast ratios in §4.1 are calculated, not tool-verified. Verify before implementation (§11.2).
- The performance budgets in §12 are set from typical figures for this class of application, not from measurement of a PBSMS build that does not yet exist. Re-baseline once stage 4 ships, in the same spirit as NFR-PERF-090.
- Timetable generation is specified as conflict detection only, consistent with SRS Chapter 17 — automatic generation remains unspecified there and is deliberately not invented here.

— *End of Document* —